

Package ‘rENM.reports’

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Type Package

Title Report generation tools for the rENM Framework

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Description Provides report generation tools for the rENM Framework, including compilation of summary tables, assembly of per-topic report pages, and production of the final species report by combining PDF products from across the rENM pipeline.

Depends R (>= 4.1.0)

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URL <https://github.com/rENM-Framework/rENM.reports>

BugReports <https://github.com/rENM-Framework/rENM.reports/issues>

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assemble_centroid_trends_page
<i>Assemble a "Centroid Trends" page</i>

Description

Builds a single-page PDF combining a suitability map with centroids and a centroid trend summary table.

Usage

```
assemble_centroid_trends_page(  
  alpha_code,  
  page_width_in = 8.5,  
  page_height_in = 11,  
  margin_top_in = 1,  
  margin_right_in = 1.75,  
  margin_bottom_in = 1,  
  margin_left_in = 1.75,  
  dpi = 300,  
  table_dpi = 600,  
  map_width_fraction = 1,  
  table_width_fraction = 1,  
  v_gap_in = 0.25  
)
```

Arguments

- alpha_code Character. Species alpha code (case-insensitive).
- page_width_in Numeric. Page width in inches. Default 8.5.
- page_height_in Numeric. Page height in inches. Default 11.
- margin_top_in Numeric. Top margin in inches. Default 1.0.
- margin_right_in Numeric. Right margin in inches. Default 1.0.
- margin_bottom_in Numeric. Bottom margin in inches. Default 1.0.
- margin_left_in Numeric. Left margin in inches. Default 1.0.

<code>dpi</code>	Integer. Working DPI for the page canvas and PNG scaling. Default 300.
<code>table_dpi</code>	Integer. Rasterization DPI for the table PDF. Default 600.
<code>map_width_fraction</code>	Numeric. Fraction (0, 1] of content width used for the top map (1.0 = full content width). Default 1.0.
<code>table_width_fraction</code>	Numeric. Fraction (0, 1] of content width used for the table (1.0 = full content width). Default 1.0.
<code>v_gap_in</code>	Numeric. Vertical gap in inches between the map and table. Default 0.25.

Details

Pipeline context This function assembles precomputed visualization outputs into a single standardized reporting page. It uses only package code and standard R functions and does not source external scripts.

Inputs Required input files:

```
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-With-Centroids.png
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Centroid-Trend-Summary.pdf
```

Outputs Output file:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Centroid-Trends.pdf
```

Logging Appends a standard eBird-format processing summary to `<project_dir>/runs/<alpha_code>/_log.txt` (blank line before dashed separator, no trailing separator).

Layout and scaling

- Top: (A) `<ALPHA>-Suitability-Trend-With-Centroids.png`
- Bottom: (B) `<ALPHA>-Centroid-Trend-Summary.pdf`
- Map scaled to `map_width_fraction` x content width
- Table scaled to `table_width_fraction` x content width
- Both preserve aspect ratio and are centered horizontally.

Methods Image composition is performed using the magick package. The summary table PDF is rasterized using `pdf_tools::pdf_convert()` at the specified DPI before scaling and placement.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if any required input is missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Appends a processing summary to the run log file.

Examples

```
## Not run:
assemble_centroid_trends_page("CASP")

## End(Not run)
```

assemble_final_report *Assemble a final report for an rENM run*

Description

Combines a predefined or user-specified set of per-page PDF products into a single report PDF for a given species run. Optionally rasterizes the pages and draws centered page numbers on all but the cover page. Optionally prepends a front matter PDF and/or appends an appendix PDF sourced from rENM.reports.

Usage

```
assemble_final_report(
  alpha_code,
  pages = NULL,
  front_matter = NULL,
  appendix = "variables.pdf",
  page_numbers = TRUE,
  dpi = 150,
  cleanup = TRUE,
  verbose = TRUE
)
```

Arguments

alpha_code	Character. Four-letter species alpha code.
pages	Character vector. Optional. Specifies which pages to include and their order. May be either: • full filenames (e.g., "BETH-Suitability-Trends.pdf"), or • short keys (see Details; e.g., "Suitability-Trends") If NULL (default), the standard page set and order are used.
front_matter	Character. Optional. Filename of a PDF located in <code>rENM.reports::inst/resources/</code> to prepend.
appendix	Character. Optional. Filename of a PDF located in <code>rENM.reports::inst/resources/</code> to append.
page_numbers	Logical. Add page numbers (default TRUE).
dpi	Numeric. Rasterization resolution when numbering.
cleanup	Logical. Remove temporary files.
verbose	Logical. Print progress messages.

Details

By default, a fixed set of pages is assembled in a predefined order. Users may override this behavior via the `pages` argument to:

- reorder pages
- omit pages
- include a subset of pages

Optional front matter and appendix PDFs must be located in `inst/resources/` within the `rENM.reports` package.

Inputs

- Pre-generated page PDFs in `<rENM_project_dir()>/runs/<alpha_code>/Summaries/pages/`
- Optional front matter / appendix PDFs in `rENM.reports::inst/resources/`

Outputs

- Final report: `<rENM_project_dir()>/runs/<alpha_code>/Summaries/ <alpha_code>-Final-Report.pdf`
- Processing summary appended to `<rENM_project_dir()>/runs/<alpha_code>/_log.txt`

Value

Character. Invisibly returns the final PDF path.

Examples

```
## Not run:
assemble_final_report("BETH")

assemble_final_report(
  "BETH",
  front_matter = "Front-Matter.pdf",
  appendix = "Appendix.pdf"
)

## End(Not run)
```

assemble_range_timeseries_page

Assemble a "Range Time Series" page

Description

Builds a single-page PDF that vertically stacks a collection of range maps above a shared caption with controlled spacing and proportional scaling.

Usage

```
assemble_range_timeseries_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_in = 1,
  target_content_width_in = 6,
  map_width_fraction = 0.85,
  caption_width_fraction = 0.85,
  gap_between_in = 0.5,
  trim_fuzz = 5,
  verbose = TRUE
)
```

Arguments

<code>alpha_code</code>	Character. Species alpha code (e.g., "CASP"). Converted to uppercase internally for file naming and paths.
<code>page_width_in</code>	Numeric. Page width in inches; default 8.5 (Letter).
<code>page_height_in</code>	Numeric. Page height in inches; default 11 (Letter).
<code>dpi</code>	Numeric. Raster resolution (dots per inch) for caption rendering and final layout. Default 300.
<code>margin_in</code>	Numeric. Uniform page margin in inches. Default 1.
<code>target_content_width_in</code>	Numeric. Target graphic width in inches when both width fractions equal 1.0. Default 6.
<code>map_width_fraction</code>	Numeric. Fraction (0,1] of <code>target_content_width_in</code> assigned to the map. Default 0.85.
<code>caption_width_fraction</code>	Numeric. Fraction (0,1] of <code>target_content_width_in</code> assigned to the caption. Default 0.85.
<code>gap_between_in</code>	Numeric. Vertical gap between map and caption in inches. Default 0.50; set to 0 for no gap.
<code>trim_fuzz</code>	Numeric. Fuzz percentage for <code>magick::image_trim()</code> . • 0 trims only perfectly uniform borders. • 5-10 often works for anti-aliased white backgrounds. Default 5.
<code>verbose</code>	Logical. If TRUE, prints diagnostic messages. Default TRUE.

Details

Pipeline context

- Input map: per-species PNG created earlier in the reporting workflow.
- Input caption: shared PDF stored in the `rENM.reports` package.
- Output page: PDF saved under each species' `Summaries/pages/` directory.

Inputs

- Range map PNG (per species): `<project_dir>/runs/<alpha_code>/Summaries/maps/<alpha_code>-Range-M`
- Range time-series caption PDF (first page only, shared): `system.file("captions", "range_timeseries_capti`
`package = "rENM.reports")`

Outputs

- Assembled PDF page: `<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Range-TimeSe`
- Log entry appended to: `<project_dir>/runs/<alpha_code>/_log.txt`

Methods

- Rasterization: `pdftools::pdf_render_page()` at the requested dpi.
- Margin trimming: `magick::image_trim()` with `trim_fuzz` tolerance.
- Scaling: width fractions applied to `target_content_width_in * dpi` pixels.
- Gap: white spacer of `gap_between_in * dpi` pixels inserted between map and caption.

Data requirements All file paths must exist and be writable; the species directory created by earlier rENM steps is assumed present.

Value

Invisible Character. Full file path of the assembled PDF.

- Structure: length-one character vector.
- Side effects: writes the PDF to disk and appends a processing summary to the species log file.

Examples

```
## Not run:
assemble_range_timeseries_page("CASP")

assemble_range_timeseries_page(
  "CASP",
  map_width_fraction      = 1.0,
  caption_width_fraction = 0.9,
  gap_between_in          = 0.25
)

## End(Not run)
```

```
assemble_state_trends_page
```

Assemble a "State Trends" page

Description

Builds a single-page PDF that combines a state-level suitability map and a state hot spot map with a summary table into a structured layout.

Usage

```

assemble_state_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
  h_gutter_in = 0.2,
  v_gap_in = 1,
  map_width_factor = 1,
  table_width_factor = 0.75,
  table_max_height_in = 2.5,
  prefer_table_pdf = TRUE,
  table_pdf_dpi = 600,
  table_png_supersample = 1.5,
  table_sharpen = TRUE,
  trim_maps = TRUE,
  trim_table = TRUE,
  trim_fuzz = 8,
  append_version_to_filename = FALSE,
  debug_frames = FALSE
)

```

Arguments

<code>alpha_code</code>	Character. Species code (case-insensitive).
<code>page_width_in</code>	Numeric. Page width in inches. Default 8.5.
<code>page_height_in</code>	Numeric. Page height in inches. Default 11.
<code>dpi</code>	Numeric. DPI for raster composition. Default 300.
<code>margin_left_in</code>	Numeric. Left margin in inches. Default 0.75.
<code>margin_right_in</code>	Numeric. Right margin in inches. Default 0.75.
<code>margin_bottom_in</code>	Numeric. Bottom margin in inches. Default 0.75.
<code>margin_top_in</code>	Numeric. Top margin in inches. Default 1.00.
<code>h_gutter_in</code>	Numeric. Horizontal gap between maps in inches. Default 0.20.
<code>v_gap_in</code>	Numeric. Vertical gap between maps and table in inches. Default 1.00.
<code>map_width_factor</code>	Numeric. Fraction (0 to 1) of content width used by the map row. Default 1.00.
<code>table_width_factor</code>	Numeric. Fraction (0 to 1) of content width used by the table. Default 0.75.
<code>table_max_height_in</code>	Numeric. Maximum table height in inches. Default 2.50.
<code>prefer_table_pdf</code>	Logical. If TRUE and a PDF exists, read the table using <code>magick::image_read_pdf()</code> . Default TRUE.

table_pdf_dpi	Integer. DPI used when rasterizing table PDF. Default 600.
table_png_supersample	Numeric. Supersampling factor for PNG tables; values greater than or equal to 1 increase resolution (for example, 2 = 200 percent). Default 1.5.
table_sharpen	Logical. If TRUE, apply mild sharpening after scaling. Default TRUE.
trim_maps	Logical. If TRUE, trim borders from map PNGs. Default TRUE.
trim_table	Logical. If TRUE, trim borders from the table image. Default TRUE.
trim_fuzz	Integer. Tolerance (0 to 100 percent) for trimming. Default 8.
append_version_to_filename	Logical. If TRUE, append a timestamp to the output filename to avoid preview caching. Default FALSE.
debug_frames	Logical. If TRUE, draw thin frames around placed images for debugging. Default FALSE.

Details

Pipeline context This function composes multiple precomputed visualization products into a single standardized reporting page. It operates entirely within the rENM project directory and does not rely on external scripts.

Inputs Inputs must exist at the following locations:

```
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-State-Analysis
<project_dir>/runs/<alpha_code>/Trends/suitability/<alpha_code>-Suitability-Trend-State-Analysis
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Suitability-Trend-Summary.png
(Optional) PDF table:
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Suitability-Trend-Summary.pdf
```

Outputs The function writes a single-page PDF to:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-State-Trends.pdf
```

Layout and composition

- Two top-aligned maps (State Analysis and Hotspots) are placed under a configurable top margin.
- The map row can be narrowed and centered using `map_width_factor`.
- A summary table is placed exactly `v_gap_in` inches below the taller of the two maps.
- The table can be narrowed and centered using `table_width_factor`.
- Layout dimensions are controlled in inches and converted to pixels using the specified DPI.

Table rendering

- If `prefer_table_pdf = TRUE`, the table is read from PDF using `magick::image_read_pdf()` and rasterized at `table_pdf_dpi` (for example, 600) to improve text clarity.
- If using a PNG, `table_png_supersample` (for example, 2 for 200 percent) increases resolution prior to fitting.
- Optional sharpening can be applied using `table_sharpen = TRUE`.

Trimming

- Uniform borders are trimmed from maps and optionally from the table to remove hidden padding.
- Trimming uses a tolerance defined by `trim_fuzz`.

Methods All image reading, trimming, scaling, and composition are handled using the `magick` package. Aspect-preserving resizing ensures consistent layout while respecting maximum width and height constraints.

Data requirements All required input files must exist prior to execution. The function will terminate with an error if required PNGs or table sources are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Emits a console message indicating the output path.

Examples

```
## Not run:
assemble_state_trends_page("CASP")

## End(Not run)
```

```
assemble_suitability_timeseries_page
      Assemble a "Suitability Time Series" page
```

Description

Builds a single-page PDF that vertically stacks a collection of suitability maps above a shared caption with controlled spacing and proportional scaling.

Usage

```
assemble_suitability_timeseries_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_in = 1,
  target_content_width_in = 6,
  map_width_fraction = 0.85,
  caption_width_fraction = 0.85,
  gap_between_in = 0.5,
  trim_fuzz = 5,
  verbose = TRUE
)
```

Arguments

alpha_code	Character. Species alpha code (for example, "CASP"). Converted internally to uppercase for file naming and paths.
page_width_in	Numeric. Page width in inches. Default 8.5.
page_height_in	Numeric. Page height in inches. Default 11.
dpi	Numeric. Target raster resolution in dots per inch for both the caption rendering and the final layout. Default 300.
margin_in	Numeric. Uniform margin on all sides in inches. Default 1.
target_content_width_in	Numeric. Target graphic width in inches for the map and caption when width fractions are 1.0. Default 6.
map_width_fraction	Numeric. Fraction (0, 1] of target width for map.
caption_width_fraction	Numeric. Fraction (0, 1] of target width for caption.
gap_between_in	Numeric. Vertical gap between map and caption in inches.
trim_fuzz	Numeric. Fuzz percentage for <code>magick::image_trim()</code> .
verbose	Logical. If TRUE, print diagnostic messages.

Details

Pipeline context For a given species alpha code (`alpha_code`), this function builds a single-page PDF that stacks:

1. The suitability contact-sheet map PNG (A) on the top of the page: `<project_dir>/runs/<alpha_code>/Summary`
2. A shared suitability time-series caption PDF (B; first page only) underneath: `system.file("captions", "suitability_timeseries_caption.pdf", package = "rENM.reports")`

Both elements are:

- Converted to raster images
- Trimmed to remove uniform white margins via `magick::image_trim()`
- Scaled so that, when `map_width_fraction = 1` and `caption_width_fraction = 1`, the visible width of both (A) and (B) is `target_content_width_in` inches (default 6), subject to the page interior width
- Separated by an adjustable vertical gap of `gap_between_in` inches
- Stacked vertically (map above gap, gap above caption)
- Placed on a white Letter-sized page (8.5 x 11 inches) with uniform margins

Trimming the source rasters first ensures that any extra baked-in white margins are removed, so the final 6-inch width corresponds to the graphic itself rather than its original page padding.

The function also appends an eBird-standard processing summary to `<project_dir>/runs/<alpha_code>/_log.txt`, including the caption source path.

Inputs

```
<project_dir>/runs/<alpha_code>/Summaries/maps/<alpha_code>-Suitability-Maps.png
system.file("captions", "suitability_timeseries_caption.pdf",
           package = "rENM.reports")
```

Outputs

<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-TimeSeries.pdf

Methods All image reading, trimming, scaling, and composition are handled using the magick package. The caption PDF is rasterized using pdftools::pdf_render_page().

Data requirements Required input files must exist prior to execution. The function will stop with an error if the map PNG or caption PDF is missing.

Value

Character. Invisibly returns the output PDF file path:

<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Suitability-TimeSeries.pdf

Side effects:

- Writes a PDF file to the project directory
- Creates output directories if they do not exist
- Appends a processing summary to the run log file

Examples

```
## Not run:
assemble_suitability_timeseries_page("CASP")

## End(Not run)
```

```
assemble_suitability_trends_page
```

Assemble a "Suitability Trends" page

Description

Builds a single-page PDF combining a suitability trend map, a suitability change trend map (acceleration/deceleration), and a caption.

Usage

```
assemble_suitability_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  margin_top_in = 1,
  margin_right_in = 1,
  margin_bottom_in = 1,
  margin_left_in = 1,
  dpi = 300,
  caption_dpi = 600,
  panel_gap_in = 0.15,
  v_gap_in = 0.25
)
```

Arguments

alpha_code	Character. Species alpha code (case-insensitive).
page_width_in	Numeric. Page width in inches. Default 8.5.
page_height_in	Numeric. Page height in inches. Default 11.
margin_top_in	Numeric. Top margin in inches. Default 1.
margin_right_in	Numeric. Right margin in inches. Default 1.
margin_bottom_in	Numeric. Bottom margin in inches. Default 1.
margin_left_in	Numeric. Left margin in inches. Default 1.
dpi	Integer. Working resolution for page/maps (dots per inch). Default 300.
caption_dpi	Integer. Rasterization DPI for the caption PDF (higher = crisper). Default 600.
panel_gap_in	Numeric. Horizontal gap (inches) between the two top panels. Default 0.15.
v_gap_in	Numeric. Vertical gap (inches) between the top row and the caption. Default 0.25.

Details

Pipeline context Produces a single-page PDF with:

- Top row:
 - (A) "-Suitability-Trend.png"
 - (B) "-Suitability-Change-Trend.png", side-by-side.
- Bottom:
 - (C) A fixed, shared caption PDF included with the rENM.reports package: "inst/captions/suitability_trend_capt" (accessed via system.file()).

The caption is rasterized via pdftools at high DPI to preserve original typography (including bold), centered horizontally, and never stretched (aspect ratio preserved).

Inputs

- Caption PDF is read directly from installed package resources (no staging).
- A and B are read as PNG files from: <project_dir>/runs/<alpha_code>/Trends/suitability/

Processing steps

- Reads the caption PDF directly from the installed package resources (no staging).
- Reads A + B (PNG) and rasterizes C (PDF) using pdftools::pdf_convert() at a specified caption_dpi (default 600).
- Scales the caption down only if wider than the content area; otherwise draws at native size, centered.
- Composites everything onto a Letter page and writes: <project_dir>/runs/<alpha_code>/Summaries/pages/"<alpha_code>-Suitability-Trends.pdf".

Outputs

- Output PDF written to: <project_dir>/runs/<alpha_code>/Summaries/pages/"<alpha_code>-Suitability-Trends.pdf"

Methods Why pdftools? Using `pdftools::pdf_convert()` (Poppler) avoids font substitution issues that can occur when ImageMagick/Ghostscript cannot find the bold font family. Rasterizing at a high DPI preserves the visual weight of bold text exactly as in the source PDF. The image is then centered and never stretched (aspect ratio preserved).

Data requirements Dependencies: Requires the magick and pdftools packages. If png is missing on your system, install it as well; `pdf_convert()` writes a PNG which is then read via magick.

Logging: Appends an eBird-standard processing summary to `<project_dir>/runs/<alpha_code>/_log.txt` with a blank line before the dashed separator and no trailing separator, including the caption source path.

Value

Character. Invisibly returns the output PDF path.

- Output file written to: `<project_dir>/runs/<alpha_code>/Summaries/pages/"<alpha_code>-Suitability-Trends.pdf"`
- Appends processing summary to: `<project_dir>/runs/<alpha_code>/_log.txt`

Examples

```
## Not run:
assemble_suitability_trends_page("CASP")
assemble_suitability_trends_page("CASP", caption_dpi = 900)

## End(Not run)
```

assemble_variable_trends_page

Assemble a "Variable Trends" page

Description

Builds a single-page PDF that vertically stacks a variable-contribution lines plot above a summary table with proportional scaling and centered alignment.

Usage

```
assemble_variable_trends_page(
  alpha_code,
  page_width_in = 8.5,
  page_height_in = 11,
  dpi = 300,
  margin_left_in = 0.75,
  margin_right_in = 0.75,
  margin_bottom_in = 0.75,
  margin_top_in = 1,
  v_gap_in = 0.5,
  plot_width_factor = 0.85,
  table_width_factor = 0.5,
  table_max_height_in = 3,
```

```

    prefer_table_pdf = TRUE,
    table_pdf_dpi = 600,
    table_png_supersample = 1.5,
    table_sharpen = TRUE,
    trim_plot = TRUE,
    trim_table = TRUE,
    trim_fuzz = 8,
    append_version_to_filename = FALSE
)

```

Arguments

alpha_code Character. Species code (case-insensitive).

page_width_in Numeric. Page width in inches. Default 8.5.

page_height_in Numeric. Page height in inches. Default 11.

dpi Numeric. DPI for the output page raster composition. Default 300.

margin_left_in Numeric. Left margin in inches. Default 0.75.

margin_right_in Numeric. Right margin in inches. Default 0.75.

margin_bottom_in Numeric. Bottom margin in inches. Default 0.75.

margin_top_in Numeric. Top margin in inches. Default 1.00.

v_gap_in Numeric. Visible gap in inches between plot and table. Default 0.50.

plot_width_factor Numeric. Fraction (0, 1] of content width used by the plot. Default 0.85.

table_width_factor Numeric. Fraction (0, 1] of content width used by the table. Default 0.50.

table_max_height_in Numeric. Maximum table height in inches. Default 3.00.

prefer_table_pdf Logical. If TRUE and a PDF exists, read the table using `magick::image_read_pdf()`. Default TRUE.

table_pdf_dpi Integer. DPI when rasterizing the table PDF. Default 600.

table_png_supersample Numeric. Greater than or equal to 1; upscale PNG tables by this factor before fitting (for example, 2 = 200 percent). Default 1.5.

table_sharpen Logical. If TRUE, apply mild sharpening after scaling. Default TRUE.

trim_plot Logical. If TRUE, trim borders from the plot image. Default TRUE.

trim_table Logical. If TRUE, trim borders from the table image. Default TRUE.

trim_fuzz Integer. Tolerance (0 to 100 percent) for trimming. Default 8.

append_version_to_filename Logical. If TRUE, append a timestamp to the output filename to avoid preview caching. Default FALSE.

Details

Pipeline context This function composes precomputed variable-trend visualizations into a single standardized reporting page. All operations are performed using package-based functionality without sourcing external scripts.

Inputs Required input files:

```
<project_dir>/runs/<alpha_code>/Trends/variables/<alpha_code>-Variable-Contributions-BR-Lines-  
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Variable-Trend-Summary.pdf  
<project_dir>/runs/<alpha_code>/Summaries/tables/<alpha_code>-Variable-Trend-Summary.png
```

Outputs Output file:

```
<project_dir>/runs/<alpha_code>/Summaries/pages/<alpha_code>-Variable-Trends.pdf
```

Layout and scaling

- Top: variable-contribution lines plot
- Bottom: summary table
- Plot is placed below the top margin
- Table is placed exactly `v_gap_in` inches below the plot
- Both elements are scaled proportionally and centered horizontally
- No cropping is performed; only scaling

Table rendering

- By default, the table is read from PDF and rasterized at high DPI using `magick::image_read_pdf()` for crisp text
- If PDF is unavailable or `prefer_table_pdf = FALSE`, a PNG is used instead
- PNG tables may be supersampled using `table_png_supersample` (for example, `2 = 200` percent)
- Optional sharpening can be applied using `table_sharpen`

Logging The function prints progress messages and appends an eBird-style processing summary to `<project_dir>/runs/<alpha_code>/_log.txt` with start and stop times and elapsed seconds. The log entry includes a blank line before the separator and no trailing separator after the entry.

Methods All image reading, trimming, scaling, and composition is handled using the `magick` package. Table rasterization from PDF uses `magick::image_read_pdf()`.

Data requirements Required input files must exist prior to execution. The function will stop with an error if required inputs are missing.

Value

Character. Invisibly returns the output PDF file path.

Side effects:

- Writes a PDF file to the project directory.
- Creates output directories if they do not exist.
- Appends a processing summary to the run log file.

Examples

```
## Not run:
assemble_variable_trends_page("CASP")

## End(Not run)
```

```
assemble_variable_trend_maps_page
```

```
    Assemble "Variable Trend Maps" pages
```

Description

Builds a multi-page PDF that shows trends and acceleration/deceleration patterns of the top contributing variables.

Usage

```
assemble_variable_trend_maps_page(alpha_code, overwrite = TRUE, verbose = TRUE)
```

Arguments

alpha_code	Character. Species alpha code (e.g., "CASP"). Converted internally to uppercase for all paths and filenames.
overwrite	Logical. If TRUE (default) overwrites existing destination files; if FALSE the function stops if a destination file already exists.
verbose	Logical. If TRUE (default) prints processing steps and paths to the console.

Details**Pipeline context**

- Copies a shared caption PDF (A) and a species-specific variable-trend-maps PDF (B) into the Summaries/pages directory for the target species.
- Ensures consistent naming so that downstream report builders pick up both pages automatically.

Inputs

- Shared caption PDF (A):
system.file("captions", "variable_trend_maps_caption.pdf", package = "rENM.reports")
- Species maps PDF (B):
file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "maps", paste0(<alpha_code>, "-Variable-Trend-Maps.pdf"))

Outputs

- Caption copy:
file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "pages", paste0(<alpha_code>, "-Variable-Trend-Maps1.pdf"))

- Maps copy:
file.path(rENM_project_dir(), "runs", <alpha_code>, "Summaries", "pages", paste0(<alpha_code>, "-Variable-Trend-Maps2.pdf"))
- Processing summary appended to:
file.path(rENM_project_dir(), "runs", <alpha_code>, "_log.txt")

Methods

- Pure file staging via file.copy() with optional overwrite.
- No PDF merging, rasterization, or resizing.

Value

A named list (invisibly returned):

- caption_page – full file path to the staged caption PDF
- maps_page – full file path to the staged maps PDF

Side effects:

- Two PDF files copied into <project>/runs/<alpha_code>/Summaries/pages
- Processing summary appended to <project>/runs/<alpha_code>/_log.txt

Examples

```
## Not run:
assemble_variable_trend_maps_page("CASP")

## End(Not run)
```

```
create_centroid_trend_summary_table
```

Create a centroid shift summary table

Description

Builds a two-part, publication-ready centroid summary for a species, combining a shift block and regression block into a standardized tabular format for reporting centroid trends. Exports to Excel and optionally to PNG and PDF.

Usage

```
create_centroid_trend_summary_table(
  alpha_code,
  decimals = 2,
  font_name = "Arial",
  title_size = 10,
  target_px = 350
)
```

Arguments

alpha_code	Character. Species alpha code (case-insensitive).
decimals	Integer. Number of decimals to display and format.
font_name	Character. Table font family.
title_size	Integer. Title font size in Excel and PNG/PDF.
target_px	Integer. Target pixel width used to center PNG tables.

Details

Pipeline context Builds a standardized centroid trend summary from precomputed centroid velocity and regression outputs for a given species.

Inputs Reads CSV files from:

- <project_dir>/runs/<ALPHA>/Trends/centroids/

Expected inputs:

Bioclimatic-Velocity <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Bioclimatic-Velocity.csv
with columns start_lon, start_lat, end_lon, end_lat, distance_km, bearing_deg, velocity_km_per_yr.

Longitude-Summary <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Centroids-Longitude-Summary.csv
with columns mean, ci_low, ci_high, pd, rope_pct.

Latitude-Summary <project_dir>/runs/<ALPHA>/Trends/centroids/ <ALPHA>-Centroids-Latitude-Summary.csv
with the same columns as Longitude-Summary.

Outputs Writes outputs to:

- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.xlsx
(always written)
- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.png
(if optional graphics packages are available)
- <project_dir>/runs/<ALPHA>/Summaries/tables/ <ALPHA>-Centroid-Trend-Summary.pdf
(if optional graphics packages are available)

Table structure

- Shift block (one row): 1980/2020 centroid coordinates, great-circle distance (km), bearing (deg), velocity (km/yr).
- Regression block (two rows: Longitude, Latitude): Slope, CI Low, CI High, PD, Rope \

Visual and typographic specification

- Title centered and bold; default 10 pt (configurable).
- Table text 8 pt (Arial default).
- No top border or rule on any block.
- No vertical lines anywhere.
- No horizontal rules between body rows in the regression block.
- Column-label rows have a light-gray ("#BFBFBF") underline; each block capped with a light-gray bottom rule.
- PNG/PDF export uses **gt** plus **webshot2** or **chromote**, with both top and bottom tables centered using table.width and table.align and canvas padding via **magick**.

- PNG/PDF use 1 px data-row padding.

Logging Prints progress to the console and appends an eBird-style processing summary (72-dash ruler; Timestamp, Alpha code, Outputs saved, Output file, Total elapsed) to: <project_dir>/runs/<ALPHA>/_log.txt.

Value

Invisibly returns a List with:

- shift_df: Data frame containing the one-row shift block.
- reg_df: Data frame containing the two-row regression block.
- files: Character vector of file paths written.

Side effects:

- Writes Excel, PNG, and PDF files to the project directory.
- Appends a processing summary to the run-specific log file.

Examples

```
## Not run:
create_centroid_trend_summary_table("CASP")
create_centroid_trend_summary_table("GRR0",
  title_size = 11, font_name = "Helvetica")

## End(Not run)
```

```
create_suitability_trend_summary_table
      Create a state-level GAP range / hot spot summary table
```

Description

Produces a publication-ready summary table from state-level suitability trend data, exporting to Excel and optionally to image and PDF formats.

Usage

```
create_suitability_trend_summary_table(alpha_code)
```

Arguments

alpha_code Character. Four-letter species code.

Details

Pipeline context Converts state-level suitability trend summary outputs into formatted tables suitable for reporting, publication, and visualization.

Inputs Input CSV must exist at:

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

Processing steps

- Reads the summary CSV using `readr::read_csv`.
- Verifies required columns are present.
- Converts numeric columns safely.
- Renames columns for presentation.
- Builds a formatted Excel workbook using `openxlsx`.
- Creates a styled table using `gt`.
- Optionally exports PNG using `webshot2` if available.
- Optionally exports PDF using `pagedown` if available.

Outputs Output files are written to:

<project_dir>/runs/<alpha_code>/Summaries/tables

- Excel (.xlsx)
- PNG (.png) if `webshot2` is available
- PDF (.pdf) if `pagedown` is available

Table columns are presented as: State, State Area, Range Area, Range \ Hot Spot Area, Hot Spot \

Log behavior Appends a processing summary to:

<project_dir>/runs/<alpha_code>/_log.txt

using the eBird-standard format.

Data requirements Input CSV must contain: state, state_area, range_area, range_pct, pos_pct, neg_pct, hotspot_area, hotspot_pct.

Value

A named list returned invisibly with the following elements:

- `xlsx`: Character. Absolute path to Excel output file.
- `png`: Character. Absolute path to PNG file or NA if skipped.
- `pdf`: Character. Absolute path to PDF file or NA if skipped.

Side effects:

- Writes formatted Excel, PNG, and PDF outputs.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
create_suitability_trend_summary_table("CASP")

## End(Not run)
```

```
create_variable_trend_summary_table
```

Create a variable trend summary table

Description

Produces a publication-ready summary table from variable-trend statistics with significance marks (*, **, ***) and bolding by PD.

Usage

```
create_variable_trend_summary_table(alpha_code)
```

Arguments

alpha_code Character. Species alpha code.

Details

Pipeline context Converts variable-level Bayesian trend statistics into a compact, formatted summary table suitable for reporting and publication.

Inputs Input file must exist at:

<project_dir>/runs/<alpha_code>/Trends/variables/ <alpha_code>-Variable-Contributions-BR-Stats.csv

Processing steps

- Reads the variable-trend CSV using readr::read_csv.
- Verifies required columns are present.
- Resolves CI column names: slope_ci_lower or slope_ci_low, and slope_ci_upper or slope_ci_high.
- Preserves row order exactly as in the input CSV.
- Assigns significance marks based on pd_slope:
 - *** if pd_slope >= 95
 - ** if 90 <= pd_slope < 95
 - * if 85 <= pd_slope < 90
- Appends significance marks to variable names.
- Converts numeric fields safely.
- Builds a formatted Excel workbook using openxlsx.
- Creates a styled table using gt.
- Optionally exports PNG using webshot2 if available.
- Optionally exports PDF using pagedown if available.

Outputs Output files are written to:

<project_dir>/runs/<alpha_code>/Summaries/tables

- Excel (.xlsx)
- PNG (.png) if webshot2 is available
- PDF (.pdf) if pagedown is available

Column mapping Output columns correspond to input columns:

- Variable <- Variable
- Points <- n_points
- Slope <- slope_mean
- CI Low <- slope_ci_lower or slope_ci_low
- CI High <- slope_ci_upper or slope_ci_high
- PD <- pd_slope
- ROPE % <- rope_slope_pct

Formatting rules

- Row order is preserved exactly as in the input CSV.
- Variables with pd_slope >= 85 are bolded.
- Significance marks are appended to variable names.

Log behavior Appends a processing summary to:

<project_dir>/runs/<alpha_code>/_log.txt

using the eBird-standard format:

- One blank line before the 72-dash separator.
- No ending separator line.
- Reports all output files written.

Data requirements Input CSV must contain: Variable, n_points, slope_mean, pd_slope, rope_slope_pct, and CI columns (slope_ci_lower or slope_ci_low, slope_ci_upper or slope_ci_high).

Value

A named list returned invisibly with the following elements:

- xlsx: Character. Absolute path to Excel output file.
- png: Character. Absolute path to PNG output or NA if skipped.
- pdf: Character. Absolute path to PDF output or NA if skipped.

Side effects:

- Writes Excel, PNG, and PDF files to disk.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
create_variable_trend_summary_table("CASP")

## End(Not run)
```

gather_range_maps	<i>Assemble 3x3 range map contact sheet</i>
-------------------	---

Description

Builds a 3x3 contact sheet of range maps for a species code by collecting PNG tiles from Time-Series outputs and arranging them on a standardized Letter-sized page layout for export to multiple formats.

Usage

```
gather_range_maps(
  alpha_code,
  margin_in = 1,
  page_width_in = 8.5,
  page_height_in = 11,
  tile = "3x3",
  trim_fuzz = 8,
  dpi = 300
)
```

Arguments

alpha_code	Character. Species code used in folder and file names.
margin_in	Numeric. Page margin on top, left, and right in inches.
page_width_in	Numeric. Page width in inches (Letter = 8.5).
page_height_in	Numeric. Page height in inches (Letter = 11).
tile	Character. Montage tile specification for magick::image_montage.
trim_fuzz	Numeric. Trim tolerance percent for magick::image_trim to remove light halos.
dpi	Numeric. Rendering resolution for raster-based layout.

Details

Pipeline context Assembles retrospective ENM range map outputs into a standardized multi-format visual summary for interpretation and reporting.

Inputs Source tiles are expected at:

<project_dir>/runs/<alpha_code>/TimeSeries/ <year>/model/ <alpha_code>-<year>-Range.png
for year in 1980, 1985, ..., 2020 (9 images total).

Processing steps

- Reads and trims each tile to remove stray white borders using magick::image_trim.
- Creates a 3x3 montage with small gutters.
- Trims the montage again to tighten outer bounds.
- Scales the montage to the content width (6.5 inches at 300 dpi).
- If the montage exceeds the content height (9 inches at 300 dpi), crops excess from the bottom while preserving the top.
- Composites onto a white Letter-sized page.

Outputs The final page is exported as:

- A one-page PNG (<alpha_code>-Range-Maps.png)
- A one-page PDF (<alpha_code>-Range-Maps.pdf)
- A one-page DOCX (<alpha_code>-Range-Maps.docx)

All files are written to:

<project_dir>/runs/<alpha_code>/Summaries/maps

The DOCX placement matches the PNG/PDF visually: the montage is scaled to the page content width (page width minus 2 x margin) and cropped from the bottom if it exceeds the content height, preserving the top of the montage so it sits 1 inch from the top margin and is centered horizontally.

Data requirements Exactly 9 range PNG files must exist for the specified alpha_code.

Value

A named list returned invisibly with the following elements:

- png: Character. Absolute file path to PNG output.
- pdf: Character. Absolute file path to PDF output.
- docx: Character. Absolute file path to DOCX output.

Side effects:

- Writes PNG, PDF, and DOCX files to the summaries directory.
- Appends a processing summary to: <project_dir>/runs/<alpha_code>/_log.txt

Examples

```
## Not run:
# For BETH using default layout and margins:
gather_range_maps("BETH")

# For a different alpha_code:
gather_range_maps("AMRO")

# Tighter trim if input PNGs have larger white borders:
gather_range_maps("BETH", trim_fuzz = 12)

## End(Not run)
```

gather_suitability_maps

Assemble side-by-side suitability maps into paged composites

Description

Builds a 3x3 contact sheet of suitability maps for a species code by collecting prediction PNGs from TimeSeries outputs and arranging them on a standardized Letter-sized page layout for export to multiple formats.

Usage

```
gather_suitability_maps(
  alpha_code,
  margin_in = 1,
  page_width_in = 8.5,
  page_height_in = 11,
  tile = "3x3",
  trim_fuzz = 8,
  dpi = 300
)
```

Arguments

alpha_code	Character. Species code used in folder and file names.
margin_in	Numeric. Page margin on top, left, and right in inches.
page_width_in	Numeric. Page width in inches (Letter = 8.5).
page_height_in	Numeric. Page height in inches (Letter = 11).
tile	Character. Montage tile specification for magick::image_montage.
trim_fuzz	Numeric. Trim tolerance percent for magick::image_trim to remove light halos.
dpi	Numeric. Rendering resolution for raster-based layout.

Details

Pipeline context Assembles retrospective ENM prediction outputs into a standardized multi-format visual summary for interpretation and reporting.

Inputs Source tiles are expected at:

<project_dir>/runs/<alpha_code>/TimeSeries/ <year>/model/ <alpha_code>-<year>-Prediction.png for year in 1980, 1985, ..., 2020 (9 images total).

Processing steps

- Reads and trims each tile to remove stray white borders using magick::image_trim.
- Creates a 3x3 montage with small gutters.
- Trims the montage again to tighten outer bounds.
- Scales the montage to the content width (6.5 inches at 300 dpi).
- If the montage exceeds the content height (9 inches at 300 dpi), crops excess from the bottom while preserving the top.
- Composites onto a white Letter-sized page.

Outputs The final page is exported as:

- A one-page PNG (<alpha_code>-Suitability-Maps.png)
- A one-page PDF (<alpha_code>-Suitability-Maps.pdf)
- A one-page DOCX (<alpha_code>-Suitability-Maps.docx)

All files are written to:

<project_dir>/runs/<alpha_code>/Summaries/maps

The DOCX placement matches the PNG/PDF visually: the montage is scaled to the page content width (page width minus 2 x margin) and cropped from the bottom if it exceeds the content height, preserving the top of the montage so it sits 1 inch from the top margin and centered horizontally.

Data requirements Exactly 9 prediction PNG files must exist for the specified alpha_code.

Value

A named list returned invisibly with the following elements:

- png: Character. Absolute file path to PNG output.
- pdf: Character. Absolute file path to PDF output.
- docx: Character. Absolute file path to DOCX output.

Side effects:

- Writes PNG, PDF, and DOCX files to the summaries directory.
- Appends a processing summary to: <project_dir>/runs/<alpha_code>/_log.txt

Examples

```
## Not run:
# For BETH using default layout and margins:
gather_suitability_maps("BETH")

# For a different alpha_code:
gather_suitability_maps("AMRO")

# Tighter trim if input PNGs have larger white borders:
gather_suitability_maps("BETH", trim_fuzz = 12)

## End(Not run)
```

```
gather_suitability_trend_stats
```

Merge state-level suitability and hot-spot statistics

Description

Reads two per-state CSV files for a species and merges them by two-letter state abbreviation to produce a concise summary table of suitability and hotspot metrics.

Usage

```
gather_suitability_trend_stats(alpha_code)
```

Arguments

alpha_code Character. Four-letter species code.

Details

Pipeline context Combines state-level suitability trend outputs with hotspot statistics into a unified summary table for downstream reporting and analysis.

Inputs Input files must exist at:

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis

Processing steps

- Reads both input CSV files using `readr::read_csv`.
- Verifies required columns are present in each dataset.
- Sorts both inputs alphabetically by state abbreviation.
- Merges datasets using a left join on state abbreviation.
- Converts relevant columns to numeric values.
- Assembles a standardized summary table with key metrics.

Outputs Output file is written to:

`<project_dir>/runs/<alpha_code>/Trends/suitability/ <alpha_code>-Suitability-Trend-State-Analysis`

Log behavior Appends a processing summary to:

`<project_dir>/runs/<alpha_code>/_log.txt`

using the eBird-standard format:

- Always skips one blank line before the 72-dash separator.
- Does not include an ending separator line.
- Omits all raster-related fields.
- Includes: Timestamp, Alpha code, Outputs saved, Total elapsed, and Output file.

Data requirements

- First file must contain: STATE, GAP.RANGE.AREA, GAP.RANGE.PCT, GAP.RANGE.POS.PCT, and either GAPP.RANGE.NEG.PCT or GAP.RANGE.NEG.PCT.
- Second file must contain: abbr, state_area_km2, hotspot_area_km2, hotspot_pct_of_state.

Value

Invisibly returns a `data.frame` with the following columns:

- `state`: Character. Two-letter state abbreviation.
- `state_area`: Numeric. State area in square kilometers.
- `range_area`: Numeric. Area of species range.
- `range_pct`: Numeric. Percent of state occupied by range.
- `pos_pct`: Numeric. Percent of positive trend area.
- `neg_pct`: Numeric. Percent of negative trend area.
- `hotspot_area`: Numeric. Area classified as hotspot.
- `hotspot_pct`: Numeric. Percent of state as hotspot.

Side effects:

- Writes a summary CSV file to the suitability directory.
- Appends a processing summary block to the run log file.

Examples

```
## Not run:
gather_suitability_trend_stats("CASP")

## End(Not run)
```

gather_top_variable_trend_maps

Assemble side-by-side variable trend maps into paged composites

Description

Pair trend and rate PNGs for each variable, build A | B montages, arrange several montages per page, and export them as PDF, DOCX, and PNG. Designed for variable-contribution reporting in the rENM workflow.

Usage

```
gather_top_variable_trend_maps(
  alpha_code,
  base_dir = c("auto", "m2", "mc"),
  base_home = "~",
  vars_csv_subpath = NULL,
  per_page = 5,
  dpi = 300,
  page_width_in = 8.5,
  page_height_in = 11,
  margin_in = 1,
  trim_fuzz = 8
)
```

Arguments

alpha_code	Character. Species code (e.g., "CASP"). Used for input and output file naming.
base_dir	Character. One of "auto", "m2", "mc"; controls which MERRA-2 directories are searched for trend images.
base_home	Character. Deprecated. Ignored and retained only for backward compatibility; the project root is taken from <code>rENM_project_dir()</code> .
vars_csv_subpath	Character or NULL. Optional path (relative to <code>runs/<alpha_code></code>) to the variable list CSV; when NULL the default path described in Pipeline steps is used.
per_page	Integer. Number of variables (rows) per output page.
dpi	Numeric. Rendering resolution in dots per inch.
page_width_in	Numeric. Page width in inches.
page_height_in	Numeric. Page height in inches.
margin_in	Numeric. Page margin in inches on all sides.
trim_fuzz	Numeric. Fuzz parameter passed to <code>magick::image_trim()</code> (in pixels).

Details

Pipeline steps

- 1. Read the variable list from `<rENM_project_dir()>/runs/<alpha_code>/Trends/variables/<alpha_code>-Variable-Contributions-BR-Stats.csv` (must contain Variable; `pd_slope` is optional).

- 2. For every variable <var> collect two PNGs (case-insensitive, punctuation-agnostic matching):
 - A (trend): <base>-trends/<var>.png
 - B (rate) : <base>-trends-rate/<var>.png

If base_dir == "auto" both "m2" and "mc" directories are tried.

- 3. Build an A | B montage for each variable with generous spacing.
- 4. Stack per_page montages per page.
- 5. Trim, down-scale, and vertically crop overflow.
- 6. Composite onto a white US Letter page (8.5 x 11 in, 1 in margins).
- 7. Export a multipage PDF, a matching DOCX, and page-level PNGs named -p01, -p02, ...

Highlight rule If the CSV includes pd_slope, any variable with {pd_slope >= 90} receives a dark-blue strip along the left edge.

Layout and styling

- Each A <-> B pair is slightly down-scaled to add whitespace.
- The gutter between A and B, as well as row spacing, is widened.
- The full column is scaled to 96\
- Excess height beyond the content area is cropped from the bottom.

CRAN compliance All hard-coded paths have been removed; the project root is obtained via rENM_project_dir().

Inputs

- Variable CSV (with optional pd_slope) as described above.
- Trend and rate PNGs residing in the two <base>-trends* folders under <rENM_project_dir()>/data/merra/.

Outputs

- <alpha_code>-Variable-Trend-Maps.pdf
- <alpha_code>-Variable-Trend-Maps.docx
- <alpha_code>-Variable-Trend-Maps-pXX.png
- A processing summary appended to runs/<alpha_code>/_log.txt

Value

A list with the following elements:

- png: Character vector of page-level PNG paths.
- pdf: Character string giving the PDF path.
- docx: Character string giving the DOCX path.

Side effects:

- Files are written to runs/<alpha_code>/Summaries/maps/.
- A log block is appended to runs/<alpha_code>/_log.txt.

Examples

```
## Not run:  
gather_top_variable_trend_maps("CASP")          # auto-detect bases  
gather_top_variable_trend_maps("CASP", "m2")     # force "m2"  
gather_top_variable_trend_maps("CASP", per_page = 3)  
  
## End(Not run)
```

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